Unit IV: Plant Anatomy (Structural Organisation)

Chapter 10

Secondary Growth

Learning Objectives

The students should be able to,

- *Analyze primary and secondary growth.*
- *Discuss the increase in length and width of the plant.*
- *Explain secondary growth in dicot stems.*
- *Explain secondary growth in dicot roots.*

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Chapter Outline

10.1 Secondary Growth in Dicot Stem

10.2 Secondary Growth in Dicot Root

How do the trees increase their girth?



Figure 10.1: *Taxus* wood

We have studied in the previous chapters the primary internal structure of monocots and dicots. If you look at the stem of grass (monocot), it is soft, whereas in the neem (dicot), the stem is very hard and woody, why? It is the secondary growth which confers the hardness to wood of dicot stems and roots. In monocots, usually there is no secondary growth and so they are soft.

The increase in girth is called **secondary growth** or **growth in girth** and we shall discuss the details of secondary growth in this chapter.

The plant organs originating from the apical meristems pass through a period of expansion in length and width. The roots and stems grow in length with the help of apical meristems. This is called **primary growth or longitudinal growth.** The gymnosperms and most angiosperms, including some monocots, show an increase in thickness of stems and roots by means of **secondary growth or latitudinal growth.**

The secondary growth in dicots and gymnosperms is brought about by two lateral meristems.

- Vascular Cambium and
- Cork Cambium

Activity

Generally monocots do not have secondary growth, but palms and bamboos have woody stems. Find the reason.

10.1 Secondary Growth in Dicot Stem

Vascular Cambium

The vascular cambium is the lateral meristem that produces the secondary

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